

Capstone Project Report

Project Title

Website Security Assessment and Incident Response Plan

1. Introduction

Cyber security is essential for protecting websites and web applications from cyber attacks. Websites are exposed to threats such as SQL Injection, Cross-Site Scripting (XSS), weak passwords, and insecure cookies. This project explains common web security risks and the importance of an effective incident response process.

2. Objective

The objective of this project is to understand common website security risks, learn the incident response process, and recommend security best practices to improve website protection.

3. Project Planning

- Study common website security threats.
 - Identify possible vulnerabilities.
 - Understand their impact on websites.
 - Prepare an incident response plan.
 - Recommend security improvements.
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4. Implementation

This project focuses on common website vulnerabilities, including:

- SQL Injection
- Cross-Site Scripting (XSS)
- Weak Passwords
- Insecure Cookies

- Missing Security Headers

These vulnerabilities can affect website security and should be addressed using secure coding and regular security assessments.

5. Incident Response Simulation

Identification

Identify the security incident and determine the affected system.

Containment

Isolate the affected system to prevent further damage.

Eradication

Remove the threat and fix the identified vulnerabilities.

Recovery

Restore normal operations and verify system security.

Lessons Learned

Document the incident and implement preventive measures to reduce future risks.

6. Security Recommendations

- Use strong passwords.
 - Validate all user input.
 - Enable HTTPS.
 - Keep software updated.
 - Enable Multi-Factor Authentication (MFA).
 - Perform regular security assessments.
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7. Conclusion

This project highlights the importance of website security and incident response. Following security best practices and responding effectively to incidents helps protect web applications from cyber attacks.

8. References

- OWASP Foundation
- NIST Cybersecurity Framework
- Cyber Security Fundamentals